Build a connected study route

Source slide - academic English

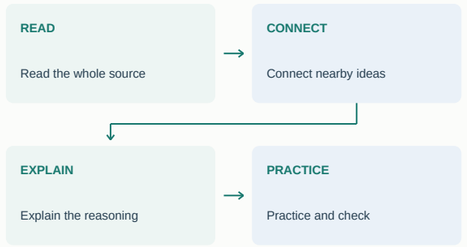
READ / CONNECT / EXPLAIN / PRACTICE

Build a connected study route

Read the complete source before explaining its parts.
Group consecutive ideas around the question they answer.

Keep the source order visible.
Name the link between each knowledge block.
Use the explanation to attempt the next task.

A route, not four unrelated summaries.



A coherent route connects the source, the reasoning and the practice.

Original neutral demonstration / Synthetic source-check counts 01 / 03

Mirror slide - plain English

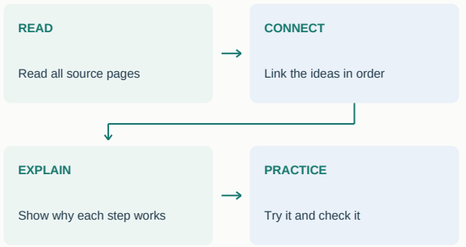
READ / CONNECT / EXPLAIN / PRACTICE

Build a connected study route

Read all of the source before teaching one small part.
Put nearby ideas together around their shared question.

Follow the order in the source.
Say how one group of ideas leads to the next.
Use what you explained to try the next task.

A route, not four unrelated summaries.



Link what the source says to why it works and how to use it.

Original neutral demonstration / Synthetic source-check counts 01 / 03

Vocabulary

Knowledge block	Consecutive source ideas answering one shared question.
Transition	The connection from a previous conclusion to the next question.
Practice	A task that uses the ideas just explained.

Prerequisite bridge: Read the whole source before teaching its parts. A formula, argument, code trace or procedure belongs in the knowledge block it helps explain. Keep the source order and make each transition explicit.

Practice connection: The complete original practice question is on study-guide page 6. Its final part asks you to distinguish a source-check record from evidence about reasoning and to choose one useful next action. (Original neutral fixture; full question page 6, solution pages 6-7.)

Original neutral demo / slide 1

Study-guide pages 1-2, 5

Check what the source log covers

Source slide - academic English

READ / CONNECT / EXPLAIN / PRACTICE

Check what the source log covers

Count distinct listed items and record which were checked.
The checked fraction describes this log, not learning quality.

Listed items: 8

Checked against the source: 6

$$q = \frac{C}{N}$$

$$q = \frac{6}{8} = \frac{3}{4}$$

Item 1 Checked	**Item 2** Checked	**Item 3** Checked	**Item 4** Checked
Item 5 Checked	**Item 6** Checked	**Item 7** Pending	**Item 8** Pending

Synthetic source log: 6 checked, 2 pending

N = listed items C = checked items q = checked fraction

The denominator defines the whole; the numerator identifies the checked part.

Original neutral demonstration / Synthetic source-check counts 02 / 03

Mirror slide - plain English

READ / CONNECT / EXPLAIN / PRACTICE

Check what the source log covers

List each item once and mark the items you have checked.
This fraction describes the list, not how well you learned.

There are 8 items on the list.

You checked 6 items against the source.

$$q = \frac{C}{N}$$

$$q = \frac{6}{8} = \frac{3}{4}$$

Item 1 Checked	**Item 2** Checked	**Item 3** Checked	**Item 4** Checked
Item 5 Checked	**Item 6** Checked	**Item 7** Pending	**Item 8** Pending

Synthetic source log: 6 checked, 2 pending

N = listed items C = checked items q = checked fraction

The bottom number is the whole list; the top number is the checked part.

Original neutral demonstration / Synthetic source-check counts 02 / 03

Vocabulary

Numerator	The top number: checked items C.
Denominator	The bottom number: all listed items N, which must be positive.
Percentage	A fraction expressed as an amount per hundred.

What the symbols and counts mean: N is the distinct listed count; C is the checked count; q is the checked fraction. The pictured log is synthetic. Its 6 of 8 checked items give 75% source-check coverage, not a score for understanding or learning quality.

Practice connection: Part (a) resets the example to N = 10 and C = 7. Use the same definition, substitute the new values, then convert the resulting share to a percentage. Do not carry over 6 and 8. (Definition page 3; complete question and guided solution pages 6-7.)

Original neutral demo / slide 2

Study-guide pages 1, 3, 6-7

Use the record to choose a next step

Source slide - academic English

READ / CONNECT / EXPLAIN / PRACTICE

Use the record to choose a next step

Separate checked items from items still awaiting a check.
Then choose a useful explanation or practice task.

$$U = N - C$$

$$r = \frac{U}{N} \qquad q + r = 1$$

A pending item calls for a source check.

A checked item can still need explanation or practice.

U = pending items r = pending fraction N > 0 C + U = N

The count tells us what was recorded; a worked explanation tests the reasoning.

Original neutral demonstration / Synthetic source-check counts 03 / 03

Mirror slide - plain English

READ / CONNECT / EXPLAIN / PRACTICE

Use the record to choose a next step

Keep checked items apart from items not yet checked.
Then pick something useful to explain or try.

$$U = N - C$$

$$r = \frac{U}{N} \qquad q + r = 1$$

Check the source for an item still marked pending.

Even a checked item may need more explanation or practice.

U = pending items r = pending fraction N > 0 C + U = N

A count shows what you logged; explain and try an idea to check your reasoning.

Original neutral demonstration / Synthetic source-check counts 03 / 03

Vocabulary

Pending	A listed item still awaiting its source check; count U.
Complement	What remains of a whole after one part is removed.
Subscript	A label identifying a particular value, such as log A or B.

Connect the relation to its conditions: The new symbol r is the pending fraction, U divided by N. The two groups cover the same nonempty list without overlap, so q + r = 1. Page 4 derives the connection one step at a time. A checked item can still need explanation or practice.

Practice connection: Parts (b)-(c) find the pending count and fraction, then check that both shares total one. Part (d) interprets the result. Page 8 compares two fresh logs and shows why larger counts and larger fractions answer different questions. (All public slides, illustrations and questions are independently authored neutral fixtures.)

Original neutral demo / slide 3

Study-guide pages 4-5, 7-8